$\mathcal{E} \rightarrow$ when storing

$\mathcal{D} \rightarrow$ when viewing back info

$\mathcal{D}$ fn $\rightarrow D(c) = P$ $\quad c \rightarrow$ cipher text

20 finding Inverse of a Matrix

① Determinant

② how to inverse

Eg $\rightarrow$ (2×2) Matrix

A $\rightarrow$ find inverse and also verify inverse using a transformation

$$A = \begin{bmatrix} 4 & 7 \\ 2 & 6 \end{bmatrix}$$

$T : R \rightarrow R$

$\Downarrow$

$A \cdot x \longrightarrow y$

$\Downarrow$

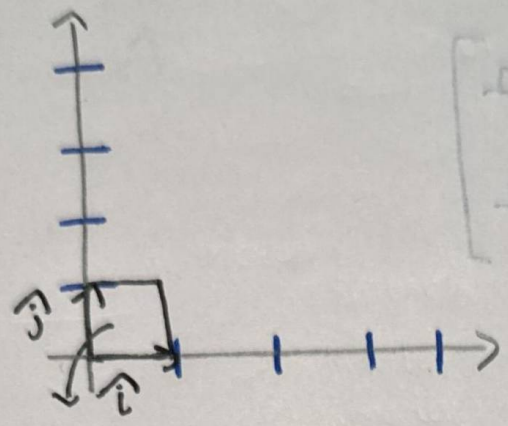
A^{-1}

inverse of A

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\boxed{A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}}$$

Determinant



Area (Specifies the determinant of $\hat{i}, \hat{j}$

Note

if det(A) = Non zero

(then defn. there is an inverse of the matrix)

Det calculation

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \rightarrow \boxed{ad - cb}$$

$$\frac{1}{24 - 14} \begin{bmatrix} 6 & -7 \\ -2 & 4 \end{bmatrix}$$

$$\frac{1}{\boxed{10}} \begin{bmatrix} 6 & -7 \\ -2 & 4 \end{bmatrix}$$

Non zero

(So, matrix A is invertible)

$$A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$A^{-1} = \frac{1}{10} \begin{bmatrix} 6 & -7 \\ -2 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 0.6 & -0.7 \\ -0.2 & 0.4 \end{bmatrix}$$

① verify using a vector

$$x = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \Rightarrow$$ transform x using A then we use A^{-1} to recover original vector

Transformation using A

$$y = Ax = \begin{bmatrix} 4 & 7 \\ 2 & 6 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 4 + 7 \\ 2 + 6 \end{bmatrix}$$

$$= \begin{bmatrix} 11 \\ 8 \end{bmatrix}$$

Recovering x using A^{-1}

$$x = A^{-1} y = \begin{bmatrix} 0.6 & -0.7 \\ -0.2 & 0.4 \end{bmatrix} \begin{bmatrix} 11 \\ 8 \end{bmatrix}$$

$$\boxed{x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}}$$